

Match the algebraic statement to its numeric example.

1.

Property	Numeric Example
Division Property of Equality	$4 + 5 + 7 = 9 + 7$
Subtraction Property of Equality	$(4 + 5) \times 7 = 9 \times 7$
Multiplication Property of Equality	$(4 + 5) \div 7 = 9 \div 7$
Symmetric Property of Equality	$4 + 5 = 9$
Addition Property of Equality	$4 + 5 - 7 = 9 - 7$

Consider the equation $33t - 99 = 0$ to answer questions 2-5.

- Complete the statement to describe the equation. The variable, t , is first being multiplied by 33, and then 99 is being subtracted from this product. The value of the expression, $33t - 99$ is equal to 0.
- Describe the operation that could be performed to isolate $33t$ on the left side of the equation.

Add 99 to both sides. (Addition Property of equality)
- Describe the operation that could be performed to isolate the t on the left side of the equation.

Divide both sides by 33 (Division Property of equality)
- Solve $-4 + \frac{x}{2} = 16$ by applying properties of equality. Explain which properties were used to solve the equation.

$$-4 + \frac{x}{2} = 16$$

$$\frac{x}{2} = 20$$

$$x = 40$$

Addition Property of Equality

Multiplication Property of Equality

6. Solve $12n + 9 = 93$ by applying properties of equality. Explain which properties were used to solve the equation.

$$\begin{array}{rcl} 12n + 9 & = & 93 \quad n = 7 \\ \underline{-9 \quad -9} & & \\ 12n & = & 84 \\ \underline{\quad \quad 12} & & \underline{\quad \quad 12} \end{array}$$

Use the equation $6x - 12 = 30$ to complete the following.

7. Apply properties of equality to solve the equation. Use substitution to verify the solution.

$$\begin{array}{rcl} 6x - 12 & = & 30 \quad x = 7 \\ +12 \quad +12 & & \text{Check:} \\ \hline 6x & = & 42 \\ \underline{\quad \quad 6} & & \underline{\quad \quad 6} \end{array} \quad \begin{array}{l} 6(7) - 12 \\ 42 - 12 = 30 \end{array}$$

Use the equation $5y + 7 = 42$ to complete the following.

8. Apply properties of equality to solve the equation. Use substitution to verify the solution.

$$\begin{array}{rcl} 5y + 7 & = & 42 \quad y = 7 \\ \underline{-7 \quad -7} & & \text{Check:} \\ 5y & = & 35 \\ \underline{\quad \quad 5} & & \underline{\quad \quad 5} \end{array} \quad \begin{array}{l} 5(7) + 7 \\ 35 + 7 = 42 \\ 42 = 42 \end{array}$$

Use the equation $\frac{x}{5} - 8 = 17$ to complete the following.

9. Apply properties of equality to solve the equation. Use substitution to verify the solution.

$$\begin{array}{rcl} \frac{x}{5} - 8 & = & 17 \\ \underline{+8 \quad +8} & & \text{Check:} \\ 5 \cdot \frac{x}{5} & = & 25 \cdot 5 \\ x & = & 125 \end{array} \quad \begin{array}{l} \frac{125}{5} - 8 \\ 25 - 8 = 17 \\ 17 = 17 \end{array}$$

Use the equation $\frac{y}{5.2} + 3.4 = 4.3$ to complete the following.

10. Apply properties of equality to solve the equation. Use substitution to verify the solution.

$$\begin{array}{rcl} \frac{y}{5.2} + 3.4 & = & 4.3 \\ \underline{-3.4 \quad -3.4} & & \text{Check:} \\ 5.2 \cdot \frac{y}{5.2} & = & 0.9 \cdot 5.2 \\ y & = & 4.68 \end{array} \quad \begin{array}{l} \frac{4.68}{5.2} + 3.4 \\ 0.9 + 3.4 = 4.3 \\ 4.3 = 4.3 \end{array}$$